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Homework

Section - 10

1

Assumptions:-

1. Many Firms sell identical products to many buyers.
2. No restrictions on entry into the market (Free entry and exit).
3. Established firms have no advantage over new ones.
4. Buyers and sellers are well informed about prices.

2

Firms are considered price takers because the market (all buyers and sellers) determines the equilibrium price and each individual firm is tiny and sells an identical product, the firm cannot charge a higher price without losing all customers, Thus the firm takes the market price; for that firm,

demand = perfectly elastic (Horizontal)

$$P = MR$$

3

long-run profits go to zero because,

Free entry and exit make profits transient:- if firms earn positive economic profits, new firms enter, market supply rises and price falls until profit is driven to zero, if firms incur losses, some exit, supply falls and price rises until losses vanish. Entry/exit stops when $P = ATC$ (zero economic profit).

4

A firm will continue to operate at a loss in the short run if it can cover its variable costs and contribute something to fixed costs. Formally:-

- Produce if $P \geq AVC$ (Price at least covers average variable cost).
- Shut down (produce 0) if $P < AVC$ (cannot cover variable cost).

if P is between AVC and ATC ($AVC \leq P < ATC$), the firm produces where $P = MC$, incurs an economic loss equal to $(ATC - P) \times Q$ but this loss is smaller than the fixed

cost it would still have to pay if it shut down.
So operating reduces the loss compared with shutting down.

5

A competitive firm maximizes profit where $P = MR = MC$
The firm's short-run supply curve is the portion of
MC above AVC.

a/ If the market price is P_3 , what quantity will the
firm produce? profit or loss?

- The firm produces where $MC = P_3 \rightarrow$ that is
 Q_3 (the vertical dropped from MC- P_3 intersection).
- At Q_3 , the ATC curve lies above P_3 (the graph
shows ATC higher than P_3).

So $P_3 < ATC \rightarrow$ the firm makes an
economic loss.

- Loss per unit = $ATC - P_3$

$$\text{Total loss} = (ATC - P_3) \times Q_3.$$

b/ Should the firm keep producing or shut down at P_3 ?

- Check Ave at Q_3 , on the graph P_3 is above Ave at Q_3 ($P_3 \geq Ave$). By Shutdown Rule —
When $P \geq Ave$ the firm should produce in the short run because it covers variable costs and reduces losses compared with shutting down.
— Produce at Q_3 , incur loss but continue operating.

c/ If the market price is P_2 , what Quantity will the firm produce? profit or loss?

- Produce where $Mc = P_2 \rightarrow$ that is Q_2 on the diagram.
- At Q_2 , $P_2 < ATe$, so producing would create an economic loss (loss per unit = $ATe - P_2$).

d/ Should the firm keep producing or shut down at P_2 ?

- Compare P_2 with Ave at the relevant output. On the provided figure P_2 is below the Ave at the Mc intersection ($P_2 < \min Ave$ region), so $P_2 < Ave$. According to Shutdown Rule — if $P < Ave$ the firm should shut down.
— Do not produce — shut down (produce 0); $\left[\begin{array}{l} \text{not limited} \\ \text{to fixed} \\ \text{costs} \end{array} \right.$